

New Time-Domain Preconditioners for HB Jacobian of RF Circuits

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Abstract—Harmonic balance (HB) method is a powerful frequency-domain method used in RF circuit simulations. The key point of HB method is efficiently solving the Jacobian system in Newton's method. In this paper, we first introduce a new time-domain preconditioner for HB Jacobian. Unlike existing time-domain preconditioners, which cannot balance the efficiency of solving the linear system corresponding to the preconditioner with the reduction in iteration step for strongly nonlinear circuit, the proposed preconditioner successfully addresses both aspects. We also present a new preconditioning method that extends time-domain preconditioners to circuit with distributed devices, which was previously unattainable. Finally, a matrix norm-based metric is proposed to measure the strength of circuit nonlinearity, which can help us a priori choose the appropriate preconditioner.

Index Terms—RF Circuit Simulation, Harmonic Balance, HB Jacobian, Preconditioning Techniques.

I. INTRODUCTION

HARMONIC balance (HB) method has been widely used in RF simulation and microwave applications [1]. HB method requires solving nonlinear equations in the frequency-domain, for which Newton's method is often used. Therefore, efficiently solving Jacobian matrix (HB Jacobian) at each Newton step is crucial in HB method. When solving HB Jacobian, Krylov subspace iterative methods such as GMRES method are widely employed. These iterative methods do not require the matrix to be formed explicitly and only require the matrix-vector product in every iteration [2], which has greatly improved the efficiency.

Despite its advantages, the effectiveness of the iterative solver for HB Jacobian depend critically on a good preconditioner which aims to reduce the iteration step. Existing preconditioning techniques for HB method can be divided into two types, the frequency and time-domain preconditioners. These two types of preconditioners each have their own applicable scenarios.

Frequency-domain preconditioner is first proposed by [3], in which the most widely used preconditioner in HB analysis, the frequency-domain average preconditioner is introduced. Several researches have also been carried out to improve the efficiency. A frequency-domain section-average preconditioner is introduced by [4], a better approximation for HB Jacobian is also established in [5] based on a one step correction strategy, and adaptive dynamical preconditioners are presented in [6] by discovering the special structure of HB Jacobian. Frequency-domain preconditioner works pretty well for circuit that is

mildly nonlinear. However, when the circuit becomes strongly nonlinear, the number of iteration step using frequency-domain preconditioner increases significantly [5] [7].

Time-domain preconditioning technique is established in [7]. Further extensions and improvements are also provided by [8] and [9]. The greatest advantage of time-domain preconditioner is its effectiveness for circuit with strong nonlinearity. Time-domain preconditioner needs to use a matrix corresponding to a low-step finite difference scheme as approximation of part of the original Jacobian matrix. Two strategies are provided by [7], which suggests using a finite difference scheme in preconditioner fully or partially. The first strategy has advantage in reducing iteration step, but solving the preconditioned system is inefficient; the second strategy's preconditioner is easier to solve, but it requires more iteration steps. Thus, existing time-domain preconditioners cannot balance the efficiency of solving the preconditioned linear system with the reduction in the number of iterations for the iterative solver. Another limitation of existing time-domain preconditioner is the incapability of handling circuit with distributed devices, while they can be directly simulated in frequency-domain [5] [7] [10].

The main contributions of our work are summarized as follows:

- We provide a new type of time-domain hybrid finite-difference preconditioner which successfully addresses both aspects mentioned above. It has significant advantages in handling strongly nonlinear circuit.
- We provide a time-domain average preconditioner, which is suitable for circuit with distributed devices.
- We introduce a matrix norm-based metric to measure the strength of circuit nonlinearity that, to some extent, can help us a priori choose the appropriate preconditioner.

The rest of the paper is organized as follows. Section II illustrates some preliminaries of the HB method and existing preconditioners. Section III introduces our newly proposed time-domain hybrid finite-difference and average preconditioner. Section IV proposes a matrix norm-based metric to measure the strength of circuit nonlinearity. Numerical results are shown in Section V, and section VI draws a conclusion.

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II. PRELIMINARIES

A. The derivation of HB Jacobian

Based on modified nodal analysis, we consider an RF circuit whose equations are given by

$$\int_{-\infty}^t y(t-s)v(s)ds + \frac{dq(v(t))}{dt} + i(v(t)) + u(t) = 0,$$

where

- $v(t) \in \mathbb{R}^n$: the voltage variables in the circuit;
- $\int_{-\infty}^t y(t-s)v(s)ds$: the distributed devices with the matrix-valued function $y(t) = 0$ for all $t < 0$;
- $q: \mathbb{R}^n \rightarrow \mathbb{R}^n$: the charge and flux storage in the circuit;
- $i: \mathbb{R}^n \rightarrow \mathbb{R}^n$: the current variables in the circuit;
- $u(t) \in \mathbb{R}^n$: the input source current waveform with period T .

Notice that the input source is T -periodic and the system is nonautonomous, the steady-state response will also be T -periodic and can be approximated as

$$v(t) \approx v_K(t) = \sum_{k=-K}^K V_k \exp(j2\pi k f_0 t), \quad f_0 = \frac{1}{T}.$$

In the following, by abuse of notation, we identify v_K as v and let $\{t_i\}_{i=0}^{2K+1}$ be the equidistant discretization of $[0, T]$. Take the approximated form of solution and apply numerical integration, we can collocate the equations in frequency-domain and write it in matrix form

$$F_{\text{HB}}(\mathcal{V}) = \mathcal{Y}\mathcal{V} + \Omega\Gamma q(\Gamma^{-1}\mathcal{V}) + \Gamma i(\Gamma^{-1}\mathcal{V}) + \mathcal{U} = 0, \quad (1)$$

where

- $\mathcal{Y} = \text{diag}\{Y_{-K}, \dots, Y_0, \dots, Y_K\}$ with Y_i 's the Fourier coefficients of $y(t)$;
- $\Omega = j2\pi \cdot \text{diag}\{0, f_0 I, \dots, K f_0 I, -K f_0 I, \dots, -f_0 I\}$;
- $\mathcal{V} = [V_0^T, V_1^T, \dots, V_K^T, V_{-K}^T, \dots, V_{-1}^T]^T \in \mathbb{C}^{n(2K+1)}$;
- Γ and Γ^{-1} are N -point FFT and inverse FFT matrices respectively.

To solve equation (1), Newton's method is used to compute $\mathcal{V}$. The k -th iteration step of Newton's method is

$$\mathcal{V}^{(k+1)} = \mathcal{V}^{(k)} - J_{\text{HB}}(\mathcal{V}^{(k)})^{-1} F_{\text{HB}}(\mathcal{V}^{(k)}).$$

For $J_{\text{HB}}(\mathcal{V}^{(k)})$, it can be computed by the chain rule and has the following form:

$$J_{\text{HB}} = \mathcal{Y} + \Omega\Gamma C\Gamma^{-1} + \Gamma\mathcal{G}\Gamma^{-1}, \quad (2)$$

where

$$\begin{aligned} C &= \text{diag} \left\{ \frac{\partial q}{\partial v} \Big|_{v=v(t_0)}, \dots, \frac{\partial q}{\partial v} \Big|_{v=v(t_{2K})} \right\} \\ &= \text{diag} \{C(t_0), \dots, C(t_{2K})\}. \end{aligned}$$

$\mathcal{G}$ can be also defined similarly.

Thus, the key point of HB method is solving the Jacobian system shown in equation (2) efficiently.

B. Frequency-Domain Preconditioner

Frequency-domain preconditioners seek sparse approximation of J_{HB} directly in the frequency-domain. Note that HB Jacobian is of the form shown in equation (2). Since $\mathcal{Y}$ and Ω are both block diagonal, the key point is to approximate $\Gamma C\Gamma^{-1}$ and $\Gamma\mathcal{G}\Gamma^{-1}$.

We take frequency-domain average preconditioner, which is the preconditioner of most use in both academic and industry field, as an example. The components of frequency-domain average preconditioner are constructed as

$$\begin{aligned} \Gamma C\Gamma^{-1} &\approx \bar{C} = I_{2K+1} \otimes \left(\frac{1}{2K+1} \sum_{i=0}^{2K} C(t_i) \right) \\ \Gamma\mathcal{G}\Gamma^{-1} &\approx \bar{\mathcal{G}} = I_{2K+1} \otimes \left(\frac{1}{2K+1} \sum_{i=0}^{2K} \mathcal{G}(t_i) \right) \end{aligned}$$

And the preconditioner $\bar{J}_{\text{HB}}$ is constructed as

$$\bar{J}_{\text{HB}} = \mathcal{Y} + \Omega\bar{C} + \bar{\mathcal{G}}.$$

It is obvious that $\bar{J}_{\text{HB}}$ is a block-diagonal matrix with $2K+1$ diagonal blocks of size $n \times n$, which means the preconditioned system $\bar{J}_{\text{HB}}z = r$ can be easily solved.

C. Time-Domain Preconditioner

1) *Circuit without Distributed Devices*: We first consider the case when no distributed devices in the circuit. Left multiply equation (2) by Γ^{-1} and right multiply by Γ , one may have

$$\Gamma^{-1}J_{\text{HB}}\Gamma = \Gamma^{-1}\Omega\Gamma C + \mathcal{G}.$$

We denote $J_t = \Gamma^{-1}J_{\text{HB}}\Gamma$ and $\mathcal{D} = \Gamma^{-1}\Omega\Gamma$, the above equation can be rewritten as

$$J_t = \mathcal{D}C + \mathcal{G}. \quad (3)$$

J_t is often called time-domain Jacobian for the diagonal blocks of C and $\mathcal{G}$ are actually variables in time-domain [7]. Time-domain preconditioning techniques seek a sparse approximation of J_t : $J_t \approx \tilde{J}_t$ and then use $\Gamma\tilde{J}_t\Gamma^{-1}$ as a preconditioner.

The special structure of $\mathcal{D}$ is the key of finding an approximation of J_t [7], it can be shown that

$$\mathcal{D} = \Gamma^{-1}\Omega\Gamma = \begin{bmatrix} 0 & \alpha_1 I_n & \alpha_2 I_n & \dots & \alpha_{-1} I_n \\ \alpha_{-1} I_n & 0 & \alpha_1 I_n & \dots & \alpha_{-2} I_n \\ & \ddots & \ddots & \ddots & \\ & & \ddots & \ddots & \alpha_1 I_n \\ \alpha_1 I_n & \alpha_2 I_n & \dots & \alpha_{-1} I_n & 0 \end{bmatrix}.$$

Assuming $\alpha_0 = 0$, it can be also proved that

$$\sum_{i=-K}^K \alpha_i = 0.$$

Thus, $\mathcal{D}C$ can be viewed as a kind of $(2K+1)$ -step finite difference scheme on a equidistant grid of $[0, T]$.

The above result suggests that $\mathcal{D}C$ can be approximated by a lower-step finite difference scheme. We refer to this kind of preconditioner more precisely as *time-domain finite difference preconditioner*:

$$J_t \approx \mathcal{D}_{\text{FD}}C + \mathcal{G},$$

here $\mathcal{D}_{\text{FD}}C$ means a lower-step finite difference scheme.

2) *Circuit with Distributed Devices*: When the circuit has distributed devices, we have time-domain Jacobian of the following form

$$J_t = \Gamma^{-1} \mathcal{Y} \Gamma + \mathcal{DC} + \mathcal{G}.$$

Unlike equation (3), there is one more term $\Gamma^{-1} \mathcal{Y} \Gamma$ in the above equation. For distributed test problems, existing techniques do not have approach to find a sparse approximation of $\Gamma^{-1} \mathcal{Y} \Gamma$, so existing time-domain preconditioners can not be applied to this kind of circuit [5] [7] [10].

3) *Two Strategies of Forming the Preconditioner*: As we have mentioned, time-domain finite difference preconditioners seek a sparse approximation of $\mathcal{DC}$ by replacing it with a finite difference scheme of lower-step. In practical, two types of strategies have been used in this process [7], we take Back-Euler scheme as an example. The first strategy retains the entire finite difference scheme:

$$\begin{aligned} \mathcal{DC} &\approx \mathcal{D}_{\text{FFD}} \mathcal{C} \\ &= \frac{1}{h} \begin{bmatrix} C(t_0) & & & -C(t_{2K}) \\ -C(t_0) & C(t_1) & & \\ & -C(t_1) & C(t_2) & \\ & & \ddots & \ddots \\ & & & -C(t_{2K-1}) & C(t_{2K}) \end{bmatrix} \end{aligned}$$

and $J_t \approx \tilde{J}_{\text{FFD}} = \mathcal{D}_{\text{FFD}} \mathcal{C} + \mathcal{G}$, we will refer to it as *time-domain full finite difference preconditioner* in subsequent sections of this paper. The second strategy eliminates the upper right corner of the finite difference scheme:

$$\begin{aligned} \mathcal{DC} &\approx \mathcal{D}_{\text{PFD}} \mathcal{C} \\ &= \frac{1}{h} \begin{bmatrix} C(t_0) & & & \\ -C(t_0) & C(t_1) & & \\ & -C(t_1) & C(t_2) & \\ & & \ddots & \ddots \\ & & & -C(t_{2K-1}) & C(t_{2K}) \end{bmatrix} \end{aligned}$$

and $J_t \approx \tilde{J}_{\text{PFD}} = \mathcal{D}_{\text{PFD}} \mathcal{C} + \mathcal{G}$. Similarly, we refer to it as *time-domain partial finite difference preconditioner*.

III. NEW TIME-DOMAIN PRECONDITIONERS

In this section we first introduce our newly proposed time-domain hybrid finite difference preconditioner. Then we present a new time-domain average preconditioning technique that can make time-domain preconditioner applicable to circuit with distributed devices.

A. Time-Domain Hybrid Finite Difference Preconditioner

We first discuss the strengths and weaknesses of the two existing time-domain preconditioning strategies mentioned in the last section. The full finite difference strategy performs very well for strongly nonlinear circuit but suffers from low efficiency in solving the preconditioned system as we have

to compute the factorization of the entire matrix [7]. The partial finite difference strategy is highly efficient in solving the preconditioned system because the system that need to be solved is block lower triangular. However, this preconditioner tends to require more iterations for circuit with strong nonlinearity. This is because the finite difference scheme used by this preconditioner corresponds to an initial value problem for an ordinary differential equation (ODE), rather than a boundary value problem for the original system.

Our newly proposed preconditioner improves upon existing strategies by incorporating the following enhancement: *using backward differencing in the reverse time-domain at the first time step*. We still use Back-Euler scheme as an example to help readers understand it:

$$\begin{aligned} \mathcal{DC} &\approx \mathcal{D}_{\text{HFD}} \mathcal{C} \\ &= \frac{1}{h} \begin{bmatrix} C(t_0) & -C(t_1) & & \\ -C(t_0) & C(t_1) & & \\ & -C(t_1) & C(t_2) & \\ & & \ddots & \ddots \\ & & & -C(t_{2K-1}) & C(t_{2K}) \end{bmatrix} \end{aligned}$$

and $J_t \approx \tilde{J}_{\text{HFD}} = \mathcal{D}_{\text{HFD}} \mathcal{C} + \mathcal{G}$. When solving the system corresponding to this preconditioner, the upper left 2×2 matrix block is solved first, followed by block forward substitution. Similar generalizations can be easily derived when using higher-step finite difference schemes such as Gear-2 scheme and we refer to this kind of preconditioner as *time-domain hybrid finite difference preconditioner*.

Combining the previous discussion on existing preconditioners, the newly proposed preconditioner has the following characteristics. First, the system that need to be solved by this preconditioner is almost block lower triangular, resulting in significantly lower computational cost compared to the time-domain full finite difference preconditioner. Second, the finite difference scheme used by this preconditioner still corresponds to a boundary value problem, providing a better approximation to the original system compared to the time-domain partial finite difference preconditioner.

B. Time-Domain Average Preconditioner

1) *Frequency Behavior of Y-parameter*: We denote the diagonal blocks of $\mathcal{Y}$ as Y_i 's, which are the Fourier coefficients of $y(t)$. In electronic engineering, the elements of matrix Y_i are called Y-parameters. Y-parameters are generally frequency-dependent, meaning they can vary with the frequency of the applied signal. In general, for linear time-invariant (LTI) circuits, Y-parameters may exhibit similar trends at integer multiples of the fundamental frequency [11].

It is known that distributed devices in RF analysis, such as transmission lines, packages and substrates are usually linear passive components. Thus, Y-parameters at integer multiples of the fundamental frequency are generally similar or related, although there might be slight variations due to the effects of impedance mismatch, losses, and other factors. This indicates that the diagonal blocks of matrix $\mathcal{Y}$ is nearly invariant.

2) *Preconditioning Procedure*: In section II we have mentioned that for system with distributed devices, the time-domain Jacobian has one more term $\Gamma^{-1}\mathcal{Y}\Gamma$, it is a dense matrix and existing preconditioning techniques have not provided a sparse approximation of it. This results in existing time-domain preconditioners being unsuitable for circuit with distributed devices.

Motivated by the behavior of Y -parameter which is discussed above, we construct a new approach which is similar to the average preconditioner in frequency-domain. The new preconditioning technique can construct a time-domain preconditioner for distributed test problems. We approximate $\Gamma^{-1}\mathcal{Y}\Gamma$ as follows:

$$\begin{aligned}\Gamma^{-1}\mathcal{Y}\Gamma &\approx \Gamma^{-1} \left(I_{2K+1} \otimes \left(\frac{1}{2K+1} \sum_{i=-K}^K Y_i \right) \right) \Gamma \\ &= (F_{2K+1}^{-1} \otimes I_N) (I_{2K+1} \otimes \bar{Y}) (F_{2K+1} \otimes I_N) \\ &= I_{2K+1} \otimes \bar{Y},\end{aligned}$$

where F_{2K+1} is a FFT matrix of order $2K+1$.

Denote $\bar{\mathcal{Y}} = I_{2K+1} \otimes \bar{Y}$ and the time-domain Jacobian J_t with distributed devices can be approximated as

$$J_t \approx \tilde{J}_t = \bar{\mathcal{Y}} + \mathcal{D}_{\text{FD}}\mathcal{C} + \mathcal{G}, \quad (4)$$

where $\mathcal{D}_{\text{FD}}\mathcal{C}$ denotes a finite difference scheme on a equidistant grid of $[0, T]$. We refer to this kind of preconditioner as *Time-domain average preconditioner*.

IV. THE METRIC OF NONLINEARITY

Previous works have mentioned that frequency-domain preconditioners are more effective for circuit with mild nonlinearity, while time-domain preconditioners are better for circuit with strong nonlinearity [5] [7]. It is natural to ask how to a priori assess the strength of circuit nonlinearity in order to select an appropriate preconditioner. In this section we introduce a matrix norm-based metric to help answer this question.

A. Block Circulant Structure in HB Jacobian

First, we have to review the special structure of HB Jacobian. The term $\Gamma\mathcal{C}\Gamma^{-1}$ and $\Gamma\mathcal{G}\Gamma^{-1}$ are both block circulant and this has been mentioned in many previous works [3] [10] [12]. Take $\Gamma\mathcal{G}\Gamma^{-1}$ as an example:

$$\Gamma\mathcal{G}\Gamma^{-1} = \begin{bmatrix} \mathcal{G}_0 & \mathcal{G}_1 & \cdots & \mathcal{G}_{2K-1} & \mathcal{G}_{2K} \\ \mathcal{G}_{2K} & \mathcal{G}_0 & \mathcal{G}_1 & & \mathcal{G}_{2K-1} \\ \vdots & \mathcal{G}_{2K} & \mathcal{G}_0 & \ddots & \vdots \\ \mathcal{G}_2 & & \ddots & \ddots & \mathcal{G}_1 \\ \mathcal{G}_1 & \mathcal{G}_2 & \cdots & \mathcal{G}_{2K} & \mathcal{G}_0 \end{bmatrix}, \quad (5)$$

where $\mathcal{G}_i$'s are all $n \times n$ block matrix and can be obtain easily by block iFFTs. $\mathcal{C}_i$'s are of the same form.

Actually, if the circuit is pure linear, which means $C(t) = \frac{\partial q}{\partial v}(t)$ and $G(t) = \frac{\partial i}{\partial v}(t)$ are both time-invariant. In this case we have

$$\mathcal{C}_i = \mathcal{G}_i = \mathbf{0}_{n \times n} \quad (i = 1, 2, \dots, 2K).$$

Thus, $\Gamma\mathcal{C}\Gamma^{-1}$ and $\Gamma\mathcal{G}\Gamma^{-1}$ are both block diagonal which means the HB Jacobian J_{HB} is block diagonal. Previous works have

also mentioned that for circuit with mild nonlinearity, J_{HB} is block diagonal dominant while the dominance will deteriorate as the nonlinearity increases [5] [10].

B. Proposed Metric of Nonlinearity

The discussion above inspires us that we can use the degree of diagonal block dominance to measure the strength of nonlinearity. Since $\mathcal{C}$ and $\mathcal{G}$ represent different circuit components, we present the following matrix norm-base metric which can measure the strength of nonlinearity of circuit components corresponding to matrix $\mathcal{C}$ and $\mathcal{G}$ respectively as following:

$$\begin{aligned}\text{LinRate}(\mathcal{C}) &= \frac{\|\mathcal{C}_0\|_{\infty}}{\sum_{i=1}^{2K} \|\mathcal{C}_i\|_{\infty}}, \\ \text{LinRate}(\mathcal{G}) &= \frac{\|\mathcal{G}_0\|_{\infty}}{\sum_{i=1}^{2K} \|\mathcal{G}_i\|_{\infty}}.\end{aligned} \quad (6)$$

The smaller the two values, the lower the dominance of the main diagonal blocks in the block circulant matrix, indicating stronger nonlinearity for the corresponding circuit components; conversely, the larger the two values, the milder the nonlinearity.

We first suppose that there is no distributed devices in the circuit. Review the form of HB Jacobian shown in equation (2), matrix $\Gamma\mathcal{C}\Gamma^{-1}$ will be left-multiplied by a block diagonal matrix Ω of coefficients resulting from differentiation. For each harmonic, the corresponding row of the block matrix will be multiplied by the coefficient $j2k\pi f_0$, which corresponds to the harmonic order. In RF simulation, the fundamental frequency f_0 is generally of a large order, typically in the range of MHz, GHz, or even higher frequencies. Thus $|j2k\pi f_0| \|\mathcal{C}_i\|$'s are often significantly larger than $\|\mathcal{G}_i\|$'s when $k \neq 0$. So for rows corresponding to non-zero order harmonics, the diagonal block dominance is almost entirely determined by $\text{LinRate}(\mathcal{C})$, while for the first row which corresponds to the zero-order harmonic it is determined by $\text{LinRate}(\mathcal{G})$. Thus, when using this metric to help test the nonlinearity of the whole circuit in order to a priori choose the appropriate preconditioner, both $\text{LinRate}(\mathcal{C})$ and $\text{LinRate}(\mathcal{G})$ have to be computed.

Besides, for circuit with distributed devices, an additional block diagonal matrix $\mathcal{Y}$ needs to be added. The elements of $\mathcal{Y}$ have different dimension compared to those of $\mathcal{C}$ and $\mathcal{G}$, and their magnitude is also generally larger. In this case, the circuit properties become more complex, making it difficult to have a universal metric to measure their nonlinear characteristics. However, numerical experiments in the next section have shown that the proposed metric can still serve to assess the degree of nonlinearity for such circuit.

TABLE I
DETAILS OF THE TEST CASES

Case	Circuit Size	Number of Harmonics	Distributed Devices
1	85	21	×
2	400	41	×
3	44	65	×
4	615	81	×
5	18	321	×
6	52	101	×
7	15	11	✓
8	15	101	✓
9	15	41	✓
10	25	21	✓

TABLE II
GMRES ITERATION STEPS OF THE PROPOSED PRECONDITIONER AND EXISTING ONES

Case	Frequency-domain Preconditioner		Time-domain Preconditioner				
	Avg		TF		TP		TH (Ours)
	Iter _{Avg}	Iter _{Avg} /Iter _{TH}	Iter _{TF}	Iter _{TF} /Iter _{TH}	Iter _{TP}	Iter _{TP} /Iter _{TH}	Iter _{TH}
1(+)	233	2.10	98	0.88	170	1.53	111
2(+)	156	1.97	58	0.73	300(*)	*	79
3(+)	300(*)	*	46	1.02	71	1.58	45
4(+)	300(*)	*	164	1.14	300(*)	*	144
5	21	0.45	42	0.91	56	1.22	46
6	67	0.69	77	0.79	300(*)	*	97

* Fails to converge after the maximum step of iterations
+ means strongly nonlinear case

TABLE III
COMPUTING TIME (TIME UNIT: SECOND)

Case	Frequency-domain Preconditioner		Time-domain Preconditioner				
	Avg		TF		TP		TH (Ours)
	T _{Avg}	T _{Avg} /T _{TH}	T _{TF}	T _{TF} /T _{TH}	T _{TP}	T _{TP} /T _{TH}	T _{TH}
1(+)	2.7821	2.14	2.4403	1.87	2.0963	1.61	1.3043
2(+)	18.6134	2.19	19.3762	2.28	23.6477(*)	*	8.4924
3(+)	3.6597(*)	*	2.0175	2.35	1.7597	2.05	0.8601
4(+)	47.3242(*)	*	70.9859	2.30	55.2421(*)	*	30.8505
5	0.5627	0.53	1.4251	1.37	1.1873	1.12	1.0659
6	1.6785	0.30	7.4829	1.33	16.3049(*)	*	5.6181

* Fails to converge after the maximum step of iterations
+ means strongly nonlinear case

V. NUMERICAL EXPERIMENTS

In this section we would like to highlight the advantage of the proposed preconditioners when dealing with strongly nonlinear circuit (circuits with or without distributed devices are both included) and the feasibility of using the proposed matrix norm-based metric to a priori assess the strength of circuit nonlinearity by numerical experiments. Also, we present a test to verify that diagonal blocks of $\mathcal{Y}$ are nearly invariant using a similar metric of the metric shown in equation (6) and this can ensure that the idea of time-domain average preconditioner is reasonable. The numerical experiments were performed in MATLAB Version 9.13.0.2320565 (R2022b) on a standard desktop with the following specifications: 12th Gen Intel(R) Core(TM) i5-12600KF 3.70 GHz 16GB RAM.

The case information is shown in Table I, including the circuit size, number of harmonics used in simulation and whether the circuits have distributed devices.

A. Time-domain Hybrid Finite Difference Preconditioner vs. Existing Preconditioners

Firstly, we compare our proposed preconditioner with existing preconditioners that all have been mentioned in previous sections on circuits without distributed devices. For frequency-domain preconditioner, we use average preconditioner. The TF, TP and TH in the table stands for time-domain full, partial and hybrid finite difference preconditioner respectively. The finite difference scheme used for approximating DC is Back-Euler scheme. Table II and III show the GMRES iteration step and computing time for different preconditioners.

We first compare our proposed preconditioner to existing time-domain preconditioners. In all the cases, the number of GMRES iteration step of our proposed preconditioner is fewer than that of the time-domain partial finite difference

preconditioner with $1.4\times$ iteration reduction on average, while the iteration step is similar to or even fewer compared to time-domain full finite difference preconditioner. For those cases in which full finite difference gain less iterations, the number is not excessively fewer. The advantage is also reflected in terms of computing time, the time-domain hybrid finite difference preconditioner has advantages over the other two time-domain preconditioners and achieves $1.9\times$ and $1.6\times$ speed up on average.

At the same time, compared to frequency-domain average preconditioner, the newly proposed preconditioner also achieves $2.0\times$ iteration reduction and $2.2\times$ computing time speed up on average in all strongly nonlinear cases (case 1-4). Even in cases where frequency-domain average preconditioner does not converge, convergence can be achieved in a few steps with our proposed preconditioner. In circuits with relatively mild nonlinearity (case 5-6), frequency-domain average preconditioner is notably superior to all time-domain preconditioners. This aligns with the experiment observation of prior related works for HB method [5] [7].

Overall, the newly proposed preconditioner shows significant advantages when dealing with circuit that exhibits strong nonlinearity.

B. Time-Domain Average Preconditioner Applied to Strongly Nonlinear Distributed Test Cases

In this part, we primarily demonstrate the value of the time-domain average preconditioner through experiments. We apply time-domain average preconditioner based on equation (4) to the four strongly nonlinear cases with distributed devices. The hybrid finite difference scheme is used to approximate DC in experiments. Table IV shows GMRES iteration step when different preconditioners are applied to these cases.

TABLE IV
TEST FOR STRONGLY NONLINEAR CIRCUITS WITH DISTRIBUTED DEVICES

Case	Frequency-domain Average Preconditioner		Time-domain Average Preconditioner
	IterFAvg	IterFAvg/IterTAvg	IterTAvg
7	37	2.18	17
8	38	1.36	28
9	155	3.04	51
10	80	3.08	26

It can be seen that in all cases, the GMRES iteration step of time-domain average preconditioner is significantly reduced compared to that of frequency-domain average preconditioner, with $2.4\times$ iteration reduction on average. The experiment confirms that the newly proposed preconditioning technique makes time-domain preconditioners applicable to circuit with distributed devices and exhibits clear advantage over frequency-domain average preconditioner when the circuit is also strongly nonlinear.

C. Preliminary Nonlinearity Test

In this subsection, we conduct a preliminary nonlinearity test on all the test cases using the aforementioned matrix norm-based metric, categorizing them into strongly nonlinear cases and mildly nonlinear cases, which is shown in Table V.

It is evident that for circuits with strong nonlinearity, at least one of the values of $\text{LinRate}(\mathcal{C})$ and $\text{LinRate}(\mathcal{G})$ will be noticeably small compared to that of circuits with mild nonlinearity (highlighted in bold in Table V). Based on our experimental observations, when $\text{LinRate}(\mathcal{C}) \lesssim 1e+13$ or $\text{LinRate}(\mathcal{G}) \lesssim 1e+02$, the circuit can be considered to have strong a priori nonlinearity; otherwise, the a priori nonlinearity is relatively weak.

From the experiments in the previous two subsections, it is also obvious that cases with mild a priori nonlinearity require fewer iterations using frequency-domain average preconditioner, while those with strong nonlinearity require fewer iterations using time-domain preconditioners. This shows that the strength of circuit nonlinearity of these cases is consistent with the a priori judgments, even when the circuit has distributed devices (case 7-10). The experiment indicates that the metric we proposed can effectively provide a priori judgments on the strength of circuit nonlinearity to a certain extent, and also help us choose the appropriate preconditioner.

TABLE V
TEST OF NONLINEARITY

Case	LinRate($\mathcal{C}$)	LinRate($\mathcal{G}$)	Consistent with the prior
Strongly Nonlinear Case			
1	4.0363e+10	4.7570	✓
2	2.2216e+13	990.4579	✓
3	1.2828e+13	26.9139	✓
4	8.8784e+09	2.7155	✓
7	4.9654e+15	1.0531	✓
8	9.8265e+14	0.9881	✓
9	1.8495e+15	0.2581	✓
10	3.2005e+15	0.9476	✓
Mildly Nonlinear Case			
5	2.2636e+14	4.1668e+03	✓
6	3.2005e+15	3.1730e+05	✓

TABLE VI
TEST OF VARIANCE OF THE DIAGONAL BLOCKS OF $\mathcal{Y}$

Case	InvarRate($\mathcal{Y}$)
7	3.2358e+07
8	2.9700e+07
9	1.7716e+09
10	1.0048e+11

D. Test of Variance of the Diagonal Blocks of $\mathcal{Y}$

In this part, we aim to show that the diagonal blocks of $\mathcal{Y}$ are nearly invariant using the same strategy of the metric shown in equation (6). It is easy to check that $\Gamma^{-1}\mathcal{Y}\Gamma$ is also block circulant. Denote the circulant blocks as $\mathcal{Y}_i$'s, and $\mathcal{Y}_i$'s can be computed similarly as $\mathcal{G}_i$'s and $\mathcal{C}_i$'s by performing block FFTs, which have been mentioned in equation (5) in section IV. Similarly, we can use the dominance of the main diagonal blocks of $\Gamma^{-1}\mathcal{Y}\Gamma$ to determine the variance of diagonal blocks of $\mathcal{Y}$. We introduce the following metric:

$$\text{InvarRate}(\mathcal{Y}) = \frac{\|\mathcal{Y}_0\|_\infty}{\sum_{i=1}^{2K} \|\mathcal{Y}_i\|_\infty}.$$

Table VI shows the values of $\text{InvarRate}(\mathcal{Y})$ for all cases with distributed devices. The values corresponding to this metric are very large, indicating that the diagonal blocks of matrix $\mathcal{Y}$ are nearly invariant. Thus, the idea of time-domain average preconditioner is reasonable.

VI. CONCLUSION

In this paper, two new time-domain preconditioners are proposed. Compared to existing time-domain preconditioners, the proposed preconditioning methods not only show significant advantages for circuit with strong nonlinearity but also enable the use of time-domain preconditioners for circuit with distributed devices. In addition, we present a matrix norm-based metric which can assist us in the preliminary selection of which preconditioner to use. In the future, we aim to conduct more in-depth mathematical research on HB-related subproblems to enhance the efficiency of the HB method for RF circuit simulation.

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